

Computational OLG with Income Uncertainty

Where we left off

The AK60 model Auerbach and Kotlikoff (1987): *deterministic* life-cycle OLG with

- $T = 60$ periods, exogenous labor productivity profile $\bar{y}^s$,
- perfect foresight over wages, returns, taxes, pensions,
- a representative household *within* each cohort.

What AK60 can and cannot say

Can: aggregate effects of pension reforms, tax changes, demographic transitions.

Cannot: *within-cohort* inequality in earnings and wealth, the role of precautionary savings, the insurance value of public pensions.

⇒ We now add **idiosyncratic risk** and **permanent heterogeneity** to the same backbone. The model becomes **AK70** (large-scale stochastic OLG).

Three motivating questions

1. What is the workhorse OLG model for studying *redistributive* policies (pensions, capital taxes, transfers) when households are heterogeneous?
2. How do we *compute* a stationary equilibrium with a non-trivial wealth distribution?
3. How well does such a model match the joint distribution of **wages**, **earnings**, and **wealth**?

Pedagogical aim

By the end of the lecture you should be able to: **(i)** write down the Bellman equation of a worker facing AR(1) productivity risk and a PAYG system; **(ii)** describe the inner $\rightarrow$ outer iteration that closes the equilibrium; **(iii)** read and modify the `AK70_stochastic_income.ipynb` notebook.

Three key facts about U.S. inequality

	U.S. Data	Target for the Model
Gini wages	0.375	0.375
Gini earnings	0.43 – 0.66	≈ 0.50
Gini wealth	0.80	?

Three different inequalities

Wages ($w \cdot \epsilon$) reflect skills/luck; **earnings** ($w \cdot \epsilon \cdot l$) add the labor-supply margin; **wealth** cumulates savings over the life cycle. A model that matches one need not match the others.

What new ingredients do we need?

A pure life-cycle model *cannot* generate within-cohort heterogeneity. We add:

- **Permanent productivity types** $e \in \{e_1, e_2\}$ (high school / college): $\rightarrow$ heterogeneity in lifetime resources.
- **Stochastic productivity** θ_t following an AR(1): $\rightarrow$ uninsurable idiosyncratic risk *à la* Aiyagari (1994).
- **Stochastic mortality** ϕ_t^s : $\rightarrow$ accidental bequests, age-dependent prudence.
- **Pay-as-you-go pensions**: $\rightarrow$ partial insurance against earnings risk and longevity risk.
- **Endogenous labor supply**: $\rightarrow$ joint determination of earnings and wealth.

This is the model in Imrohoroglu et al (1995), Hugget (1996).

Demographics

Demographic structure

- Each period a new cohort of equal measure is born at *model age* $s = 1$ (real-life age 21).
- Maximum lifespan: $T = 70$ model periods (real-life 21–90).
- Survival from age s to $s + 1$ is stochastic with probability ϕ_t^s .
- Working life $T^W = 45$ periods (real-life 21–65); retirement is mandatory thereafter.
Maximum retirement length: $T^R = T - T^W = 25$ periods.

Cohort sizes

Let $N_t(s)$ denote the number of s -year-olds at t . Population $N_t = \sum_s N_t(s)$. The newborn cohort grows at rate n_t :

$$N_{t+1}(1) = (1 + n_t) N_t(1).$$

In a stationary equilibrium with ϕ^s constant, $g_n = n$ and the cohort shares $\mu^s = N_t(s)/N_t$ are time-invariant.

Stationary cohort shares

Iterate forward from μ^1 :

$$\mu^{s+1} = \frac{\phi^s}{1+n} \mu^s, \quad s = 1, \dots, T-1, \quad (1)$$

then normalise so that $\sum_{s=1}^T \mu^s = 1$.

Calibration: survival probabilities from a UN/Bureau of the Census life table; $n = 0.754\%$ to match U.S. demographics.

Why this matters numerically

The mortality discount $\phi^s/(1+n)$ enters *both* the household's expected utility *and* the law of motion for the wealth distribution. Inconsistency between the two is a common source of bugs: the same ϕ^s must appear in both places.

Households

Preferences: balanced-growth-consistent CRRA

A newborn agent maximises expected lifetime utility:

$$\max \sum_{s=1}^T \beta^{s-1} \left(\prod_{j=1}^s \phi_{t+j-1}^{j-1} \right) \mathbb{E}_t \left[u(c_{t+s-1}^s, 1 - l_{t+s-1}^s) + v(g_{t+s-1}) \right], \quad (2)$$

with instantaneous utility

$$u(c, 1 - l) = \frac{(c^\gamma (1 - l)^{1-\gamma})^{1-\eta}}{1 - \eta}. \quad (3)$$

Parameters: β discount factor, $1/\eta$ IES, γ consumption share in utility, $v(g)$ additive utility from public goods (irrelevant for the household problem).

King–Plosser–Rebelo restriction

The functional form is the unique CRRA with leisure consistent with balanced growth when wages grow at rate g_A . This is what makes the stationary transformation later possible.

Productivity: three sources of heterogeneity

Idiosyncratic labour productivity:

$$\epsilon(s, \theta, e) = \theta \cdot e \cdot \bar{y}^s, \quad (4)$$

- $\bar{y}^s$ **age component** (deterministic hump), calibrated from Hansen (1993).
- $e \in \{0.57, 1.43\}$ **permanent type** (high school vs. college), $\pi(e_j) = 1/2$ – delivers a 150% college wage premium.
- θ **stochastic AR(1)** component:

$$\ln \theta' = \rho \ln \theta + \xi, \quad \xi \sim \mathcal{N}(0, \sigma_\xi^2). \quad (5)$$

Decomposition of wage variance

$\text{Var}(\ln w) = \underbrace{\text{Var}(\ln \bar{y}^s)}_{\text{age}} + \underbrace{\text{Var}(\ln e)}_{\text{permanent}} + \underbrace{\text{Var}(\ln \theta)}_{\text{transitory}}$. The permanent component is responsible for most lifetime inequality; the AR(1) drives within-cohort dispersion that grows with age.

Budget constraint of the household

Net labour income:

$$y_t^s = \begin{cases} (1 - \tau_t^l - \tau_t^p) \epsilon(s, \theta, e) A_t w_t l_t^s, & s \leq T^W, \\ pen_t, & s > T^W, \end{cases} \quad (6)$$

with A_t aggregate productivity ($g_A = 2\%$).

Period budget at age s :

$$k_{t+1}^{s+1} + b_{t+1}^{s+1} + (1 + \tau_t^c) c_t^s = y_t^s + tr_t + [1 + (1 - \tau_t^k)(r_t - \delta)] k_t^s + (1 + r_t^b) b_t^s. \quad (7)$$

$a_t^s = k_t^s + b_t^s$ assets; $\tau^l, \tau^p, \tau^k, \tau^c$ labour, pension, capital, consumption tax rates; tr_t government transfers.

The household's recursive problem

Individual state: $z_s = (\theta, e, a)$ for workers; $z_s = (a)$ for retirees.

Bellman equation

$$V_t(z_s) = \max_{c, l, a'} \left\{ u(c, 1 - l) + v(g) + \beta \phi_t^s \sum_{\theta'} \text{Prob}(\theta' | \theta) V_{t+1}(z_{s+1}) \right\}, \quad (8)$$

subject to (7) and the borrowing constraint $a' \geq 0$. Terminal condition: $V_t(z_{T+1}) = 0$.

- For **retirees** ($s > T^W$): $l = 0$, only a' is chosen.
- For **workers**: jointly choose a' and l ; with CRRA-Cobb-Douglas utility the FOC for l admits a closed form given a' (*nested* problem).
- The credit constraint $a' \geq 0$ binds at the bottom of the wealth distribution and is precisely the Aiyagari (1994) precautionary-saving channel.

Firms, Government, and Equilibrium

A representative firm with Cobb–Douglas technology

$$Y_t = F(K_t, A_t L_t) = K_t^\alpha (A_t L_t)^{1-\alpha}, \quad (9)$$

hires capital and effective labour at competitive prices:

$$w_t = (1 - \alpha) K_t^\alpha (A_t L_t)^{-\alpha}, \quad r_t = \alpha K_t^{\alpha-1} (A_t L_t)^{1-\alpha}. \quad (10)$$

Aggregate effective labour:

$$L_t = \sum_{s=1}^{T^w} \mu^s \int \epsilon(s, \theta, e) l(s, \theta, e, a) dF(\theta, e, a; s). \quad (11)$$

Two interest rates, one return

On the household side, capital and government bonds yield $(1 - \tau^k)(r - \delta)$ and r^b respectively.

No-arbitrage pins them down equal $(1 - \tau^k)(r - \delta) = r^b$.

Government budget:

$$Tr_t + G_t + r_t^b B_t = Tax_t + Beq_t + (B_{t+1} - B_t), \quad (12)$$

with tax revenue

$$Tax_t = \tau_t^l A_t L_t w_t + \tau_t^k (r_t - \delta) K_t + \tau_t^c C_t. \quad (13)$$

Balanced PAYG pension system:

$$Pen_t = \tau_t^p A_t L_t w_t, \quad (14)$$

with replacement rate *repl* defining $pen_t = repl \cdot w_t \bar{l}$. The contribution rate τ_t^p *adjusts* to balance the system in equilibrium.

Accidental bequests

Households who die between t and $t + 1$ leave their assets to the government, which redistributes them as part of Tr_t . This is the simplest closure.

Stationary transformation

Detrend by A_t and per-capita by N_t :

$$\tilde{x}_t^s \equiv \frac{x_t^s}{A_t}, \quad \tilde{X}_t \equiv \frac{X_t}{A_t N_t}, \quad \tilde{L}_t \equiv \frac{L_t}{N_t}. \quad (15)$$

The household budget in stationary form:

$$(1 + \tau^c) \tilde{c} = \tilde{y} + [1 + (1 - \tau^k)(r - \delta)] \tilde{a} + \tilde{t}r - (1 + g_A) \tilde{a}'. \quad (16)$$

The Bellman equation acquires a growth-adjusted discount factor:

$$V(\tilde{z}_s) = \max_{\tilde{c}, l, \tilde{a}'} \left\{ u(\tilde{c}, 1 - l) + (1 + g_A)^{\gamma(1-\eta)} \beta \phi^s \sum_{\theta'} \text{Prob}(\theta' | \theta) V(\tilde{z}_{s+1}) \right\}. \quad (17)$$

Why $(1 + g_A)^{\gamma(1-\eta)}$ appears

Stationarising $u(c, 1 - l)$ pulls out a factor $A_t^{\gamma(1-\eta)}$ in front of the continuation value. With $\gamma = 0.33$, $\eta = 2$ this exponent is *negative*, so growth *reduces* the effective discount factor.

Stationary recursive equilibrium

A list $\{\tilde{a}'(z), \tilde{c}(z), l(z); \tilde{K}, \tilde{L}, \tilde{B}, \tilde{C}, \tilde{\Omega}, w, r, r^b, \tau^p, \tilde{T}r, f(z)\}$ such that:

1. Policy functions solve the stationary Bellman equation.
2. Firms' first-order conditions hold.
3. Capital market: $\tilde{\Omega} = \tilde{K} + \tilde{B}$.
4. No-arbitrage: $r^b = (1 - \tau^k)(r - \delta)$.
5. Government and PAYG budgets are balanced.
6. Goods market: $\tilde{Y} = \tilde{C} + \tilde{G} + (1 + g_A)(1 + n)\tilde{K}' - (1 - \delta)\tilde{K}$.
7. The cross-sectional measure $f(z)$ is a fixed point of the law of motion induced by $\tilde{a}'(\cdot)$, mortality ϕ^s , growth n , and $\text{Prob}(\theta'|\theta)$.

Calibration

Parameters

Parameter	Value	Description
α	0.35	capital share in production
δ	8.3%	depreciation rate
g_A	2.0%	TFP growth
$1/\eta$	0.5	intertemporal elasticity of substitution
γ	0.33	consumption weight in utility
β	1.011	discount factor
n	0.754%	population growth rate
$\tau^l + \tau^p$	28%	labour-income wedge
τ^k	36%	capital-income tax rate
τ^c	5%	consumption tax rate
G/Y	18%	government consumption-output ratio
B/Y	63%	debt-output ratio
$repl$	35.2%	gross pension replacement rate
$\{e_1, e_2\}$	$\{0.57, 1.43\}$	permanent productivity types

$\beta > 1$ is consistent with discounting once mortality and growth adjustments are applied: the *effective* discount factor $\beta(1 + g_A)^{\gamma(1-\eta)}\phi^s$ is well below one.

Calibrating the AR(1) productivity process

Following Huggett (1996): $\rho = 0.96$, $\sigma_{\xi}^2 = 0.045$.

We discretise $\ln \theta$ with $n_{\theta} = 5$ equally-spaced grid points on $[-m\sigma_{y_1}, m\sigma_{y_1}]$ with $m = 1$ and $\sigma_{y_1} = 0.38$:

$$\Theta = (0.4688, 0.6847, 1.0000, 1.4605, 2.1332),$$

chosen so that the Gini of hourly wages matches 0.374 in the data.

Tauchen's method

The Markov transition matrix $\text{Prob}(\theta'|\theta)$ is constructed by integrating the conditional Gaussian density of ξ over the bins around each grid point (Tauchen, 1986):

$$\text{Prob}(\theta'|\theta) = \begin{pmatrix} 0.7734 & 0.2210 & 0.0056 & 0.0000 & 0.0000 \\ 0.1675 & 0.6268 & 0.2011 & 0.0046 & 0.0000 \\ 0.0037 & 0.1823 & 0.6281 & 0.1823 & 0.0033 \\ 0.0000 & 0.0046 & 0.2011 & 0.6268 & 0.1675 \\ 0.0000 & 0.0000 & 0.0056 & 0.2210 & 0.7734 \end{pmatrix}.$$

Initial cohort distribution

For each permanent type e_j , the initial distribution over θ matches the area under the lognormal density:

$$\nu(\theta) = (0.1783, 0.2010, 0.2413, 0.2010, 0.1783)^\top.$$

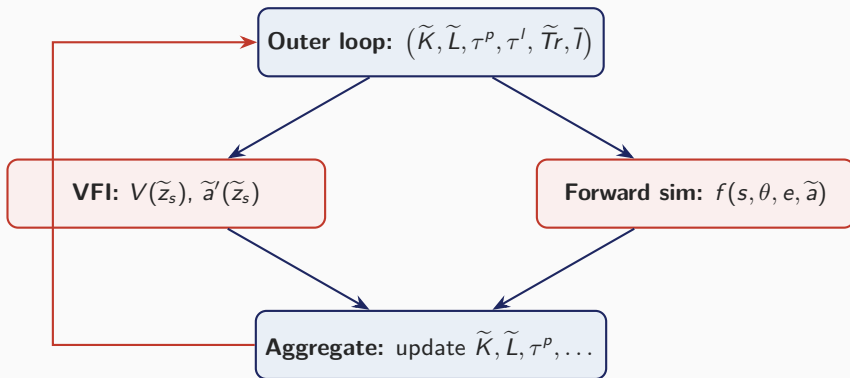
The age efficiency profile $\bar{y}^s$ is hump-shaped, peaking around real-life age 50, calibrated to Hansen (1993).

Why a 5-state grid is enough

Increasing n_θ has decreasing returns once Tauchen's grid covers $\pm m\sigma$ with $m \geq 1$. The bottleneck for the wealth distribution is the *distribution grid* n_{ag} , not the number of productivity states.

Computing the Stationary Equilibrium

Architecture: nested loops



- **Inner block:** backward induction (VFI) + forward simulation of measure.
- **Outer block:** fixed-point iteration on aggregate prices and policies.
- Convergence: linear updating with damping factor $\lambda \approx 0.2$.

The 8-step algorithm

1. Parameterise the model and choose grids on the asset space.
2. Initial guesses for $\tilde{K}$, $\tilde{L}$, $\tilde{I}$, τ^P , τ^I , $\tilde{T}r$.
3. Compute factor prices w , r and pension benefit $\widetilde{pen}$.
4. Solve the household problem by backward induction (VFI).
5. Simulate the cross-sectional measure $f(s, \theta, e, \tilde{a})$ forward.
6. Compute $\tilde{\Omega}$, $\tilde{L}$, $\tilde{I}$, $\widetilde{Tax}$, $\tilde{T}r$.
7. Update aggregate variables; iterate from Step 3 until convergence.
8. If needed, refine the asset grid and restart.

Two grids, two roles

$n_a = 500$ for the *policy function* (interpolated when evaluating expectations); $n_{ag} = 1,000$ for the *distribution* (allocated by linear weights $\rightarrow$ preserves total mass exactly).

Step 1–2: grids and initial guess

Asset grids. Equispaced on $[\tilde{a}^{\min}, \tilde{a}^{\max}]$ with $\tilde{a}^{\min} = 0$ (borrowing constraint). After convergence, check that $f(\tilde{a}^{\max})$ is small (else widen).

Initial guess (interior steady state).

- $\bar{l} = 0.30$
- $\tilde{L} = 0.30$ (working-age share $\approx 78\%$)
- $r = 3\% \Rightarrow \tilde{K} \approx 1.708$
- $\tilde{\Omega} \approx 1.26 \tilde{K}$ (since $B/Y = 0.63$, $K/Y \approx 3$)
- $\tilde{Tr} = 0.01$

Bad initial conditions kill convergence

With non-convex policy functions induced by the borrowing constraint, the outer loop is not globally contractive. Starting from a point too far from equilibrium can produce oscillations between regions where the constraint is/isn't binding.

Step 3: prices in stationary form

Given the aggregate guesses:

$$w = (1 - \alpha)\tilde{K}^\alpha \tilde{L}^{-\alpha}, \quad r = \alpha \tilde{K}^{\alpha-1} \tilde{L}^{1-\alpha}, \quad (18)$$

$$r^b = (1 - \tau^k)(r - \delta), \quad \widetilde{pen} = repl \cdot w \bar{l}. \quad (19)$$

What pins r^b down

The bond return is *not* an independent equilibrium object once r is determined: arbitrage forces $r^b = (1 - \tau^k)(r - \delta)$. The model has *one* price of capital, two financial assets that look the same to households after taxes.

Step 4: VFI by backward induction

Solve from the terminal age $s = T$ backwards.

Last period ($s = T$): no choice, terminal consumption from budget,

$$\tilde{c}^T = \frac{\widetilde{pen} + [1 + (1 - \tau^k)(r - \delta)]\tilde{a}^T + \widetilde{Tr}}{1 + \tau^c}, \quad V^T(\tilde{a}^T) = u(\tilde{c}^T, 1). \quad (20)$$

Retiree ages $s = T^W + 1, \dots, T - 1$: 1-D max in $\tilde{a}'$ via Golden Section Search.

Worker ages $s = T^W, \dots, 1$: 2-D problem in $(\tilde{a}', l)$. Use the *intra-temporal* FOC to eliminate l given $\tilde{a}'$:

$$l = \gamma - \frac{1 - \gamma}{(1 - \tau^l - \tau^p) \epsilon(s, \theta, e)_w} \left([1 + (1 - \tau^k)(r - \delta)]\tilde{a} + \widetilde{Tr} - (1 + g_A)\tilde{a}' \right). \quad (21)$$

$\Rightarrow$ *Nested* 1-D optimisation in $\tilde{a}'$.

Step 4: interpolation and accuracy check

Interpolation. The next-period continuation value $V_{t+1}(\tilde{a}', \theta', e)$ is evaluated at off-grid $\tilde{a}'$ using **linear** or **cubic** splines. Linear is monotone and robust; cubic is smoother but can introduce non-monotonicity in policy functions.

Euler-equation residual

For an s -year-old worker:

$$R(\tilde{a}) = 1 - \frac{u_c(\tilde{c}, 1 - l)}{\beta (1 + r^b) (1 + g_A)^{\gamma(1-\eta)-1} \phi^s \mathbb{E} u_c(\tilde{c}', 1 - l')}. \quad (22)$$

We average $|R|$ over all grid points (or weight by the measure $f(\cdot)$). **Targets:** max residual $\leq 10^{-3}$, mean residual $\leq 10^{-4}$.

Step 5: simulating the cross-sectional measure

The measure $f(s, \theta, e, \tilde{a})$ lives on $n_{ag} \times n_e \times n_\theta \times T^W + n_{ag} \times T^R \approx 4.75 \times 10^5$ grid points.

Initialisation at age $s = 1$:

$$f(1, \theta, e, \tilde{a}) = \begin{cases} \frac{1}{2} \mu^1 \nu(\theta) & \text{if } \tilde{a} = 0, \\ 0 & \text{otherwise.} \end{cases}$$

Forward update. For each $(s, \theta, e, \tilde{a}_i)$:

1. Read off $\tilde{a}' = \tilde{a}'(s, \theta, e, \tilde{a}_i)$.
2. Allocate $\tilde{a}'$ to its two neighbouring grid points by linear weights (preserves total mass).
3. Multiply by $\text{Prob}(\theta'|\theta)$ to spread over next-period productivity states.
4. Apply the survival adjustment $\phi^s/(1+n)$.

Step 6: aggregation and budget closures

With $f(\cdot)$ in hand:

$$\begin{aligned}\tilde{\Omega} &= \sum_{s, \theta, e, \tilde{a}} \tilde{a} f(s, \theta, e, \tilde{a}), \\ \tilde{L} &= \sum_{s \leq T^W, \theta, e, \tilde{a}} \epsilon(s, \theta, e) l(s, \theta, e, \tilde{a}) f(\cdot), \\ \tilde{C} &= \sum_{s, \theta, e, \tilde{a}} \tilde{c}(s, \theta, e, \tilde{a}) f(\cdot).\end{aligned}$$

Then $\tilde{K} = \tilde{\Omega} - \tilde{B}$ closes the capital market.

Pension contribution rate (from PAYG balance):

$$\tau^P = \frac{\widetilde{Pen}}{w \tilde{L}}. \quad (23)$$

Government transfers (residual from gov't budget):

$$\widetilde{Tr} = \widetilde{Tax} + [(1 + g_A)(1 + n) - (1 + r^b)]\tilde{B} - \tilde{G}. \quad (24) \quad 26/32$$

Step 7–8: outer loop and grid refinement

Damped fixed-point update: for each aggregate $X \in \{\tilde{K}, \tilde{L}, \tau^p, \tau^l, \tilde{Tr}, \tilde{l}\}$:

$$X^{\text{new}} = (1 - \lambda) X^{\text{old}} + \lambda X^{\text{implied}}, \quad \lambda \approx 0.2. \quad (25)$$

Iterate until $\|X^{\text{new}} - X^{\text{old}}\|_{\infty} < \text{tol}$ (e.g. 10^{-4}).

Grid diagnostics (Step 8):

- Is the right tail of $f(\tilde{a})$ negligible at $\tilde{a}^{\max}$? If not, widen the grid.
- Are Euler residuals satisfactory? If not, increase n_a or switch interpolant.
- Does $\sum f = 1$ hold to machine precision after each iteration?

Damping vs. Anderson acceleration

Linear damping with $\lambda = 0.2$ is slow but robust. Anderson acceleration can cut runtime by a factor of 5–10 but may diverge near non-smooth regions of the mapping; use it only after a basic version is working.

Results

[Insert figure: wealth–age profile $\tilde{a}^s$]

- Wealth peaks around real-life age 65, declines thereafter (retirement dissaving).
- Permanent type e_2 accumulates substantially more.
- Within-cohort variance of wealth grows monotonically with s until retirement, then is compressed by mortality.

[Insert figures: consumption profile $\tilde{c}^s$ and labour profile l^s]

- Consumption tracks the hump in efficiency $\bar{y}^s$ less than earnings, consistent with smoothing motives.
- Labour supply falls toward retirement as the wedge between marginal utility of leisure and consumption rises.
- Mortality risk steepens the consumption profile relative to the AK60 deterministic benchmark.

Lorenz curves: earnings and wealth

[Insert figure: Lorenz curves of earnings and wealth, model vs. U.S. data]

How we score

	U.S. Data	AK70 Model
Gini wages	0.375	0.375
Gini earnings	0.43 – 0.66	0.505
Gini wealth	0.80	0.66

What the model misses on wealth inequality

The AK70 model already explains the bulk of *earnings* inequality but **understates wealth concentration** (especially the top 1%). Quantitative literature has filled the gap with five extensions:

1. **Bequest motives** – De Nardi, Yang (2016).
2. **Unemployment risk** – Heer (2003).
3. **Asset-tested transfers** – Hubbard Skinner Zeldes (1995).
4. **Entrepreneurship** – Cagetti de Nardi 2009.
5. **Stochastic health and medical expenses** – Jung Tran (2016).
6. **Family heterogeneity (children, marriage)** – Holter Krueger Stepanchuk (2019).

Bottom-line for policy

With realistic Gini values for earnings, the AK70 framework is well suited to study *redistributive* pension reforms. For very-top-of-the-distribution wealth questions (estate taxation, billionaire wealth), one of the extensions above is essential.

Wrap-up

What we did today

- Extended the **deterministic AK60** to a **stochastic large-scale OLG** with permanent and transitory productivity heterogeneity, endogenous labour, mortality risk, PAYG pensions, and government debt.
- Set up the household's recursive problem and the stationary equilibrium as a *fixed point* in aggregates and a cross-sectional measure.
- Walked through an **8-step algorithm** that closes the equilibrium with a nested VFI + forward simulation block inside an outer fixed-point loop.
- Compared model-implied Gini coefficients against U.S. targets and identified the main sources of unexplained wealth concentration.

Lab session

We will run `AK70_stochastic_income.ipynb` on Colab, inspect Euler residuals as a diagnostic, and explore comparative statics with respect to the replacement rate *repl*.